

Traffic Safety and Fatal Motor Vehicle Crashes

[Github Link](#)

1. Policy Problem, Audience, and Decision

Motor vehicle crashes remain an important public policy and transportation safety issue because fatal crashes result in preventable loss of life and impose substantial social and economic costs. For transportation agencies, understanding when and where fatal crashes occur can help identify patterns that may warrant additional enforcement, roadway interventions, public education, or other traffic-safety strategies. This project focuses specifically on fatal motor vehicle traffic crashes, rather than crashes of all severities.

The audience will be the State Department of Transportation (DOT) or transportation safety agency, with transportation planners and traffic-safety policymakers as the primary audience. The final portfolio will provide an evidence-based overview of when and where fatal crashes occur and how these patterns have changed over time.

This project is intended to support situational understanding and prioritization, rather than identify causal relationships. Specifically, the analysis can help transportation safety professionals identify periods and settings in which fatal crashes are concentrated and determine whether those patterns have changed over time. The project will address three focused analytical questions:

1. When do fatal crashes occur most frequently during the week?
2. How does the burden of fatal crashes vary across states after accounting for population size?
3. How have the timing patterns of fatal crashes changed from 2020 to 2024?

Together, these questions examine the when, where, and change over time of fatal crashes.

2. Data Viability	
Data Source	National Highway Traffic Safety Administration (NHTSA) U.S. Census Bureau
Dataset	National FARS dataset 2024 State Population. NST-EST2024-POP.xlsx
Unit of observation	One fatal motor vehicle traffic crash (FARS); state population estimate (Census)
Population represented	Census of qualifying fatal motor vehicle traffic crashes
Geographic coverage	50 states and DC
Time coverage used	2020–2024
2024 size	36,297 observations × 80 variables in accident.csv
Key variables	ST_CASE, DAY_WEEKNAME, HOUR, HOURNAME, RUR_URBNAME, YEAR, FATALS
Why variables matter	Identify crash timing, urban/rural location, annual patterns, and fatalities

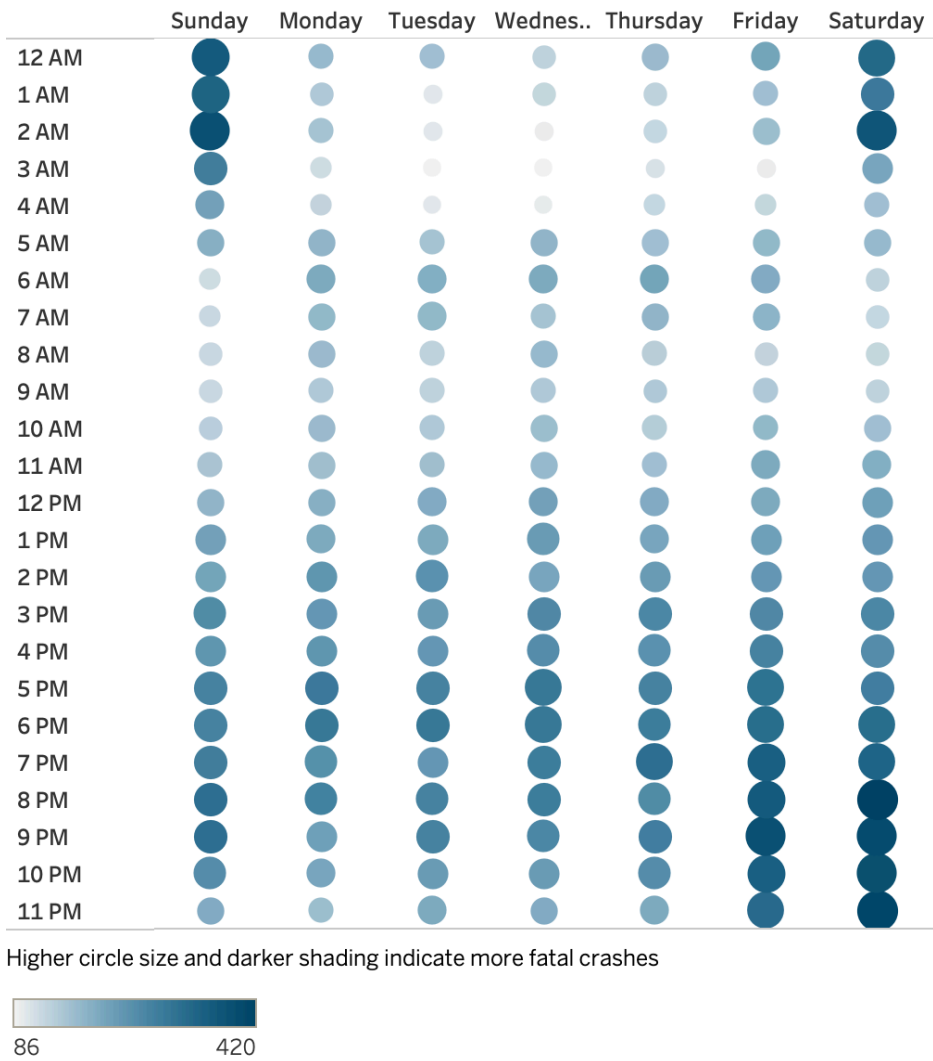
The data has limitations that some variables contain unknown or not-reported values; for example, HOUR = 99 represents unknown hours. Also, FARS only represents fatal crashes, so

counts cannot be interpreted as overall crash risk without a denominator such as total crashes or vehicle miles traveled.

3. Proof-of-Concept Visualizations

Prototype 1: When do fatal crashes occur most frequently during the week?

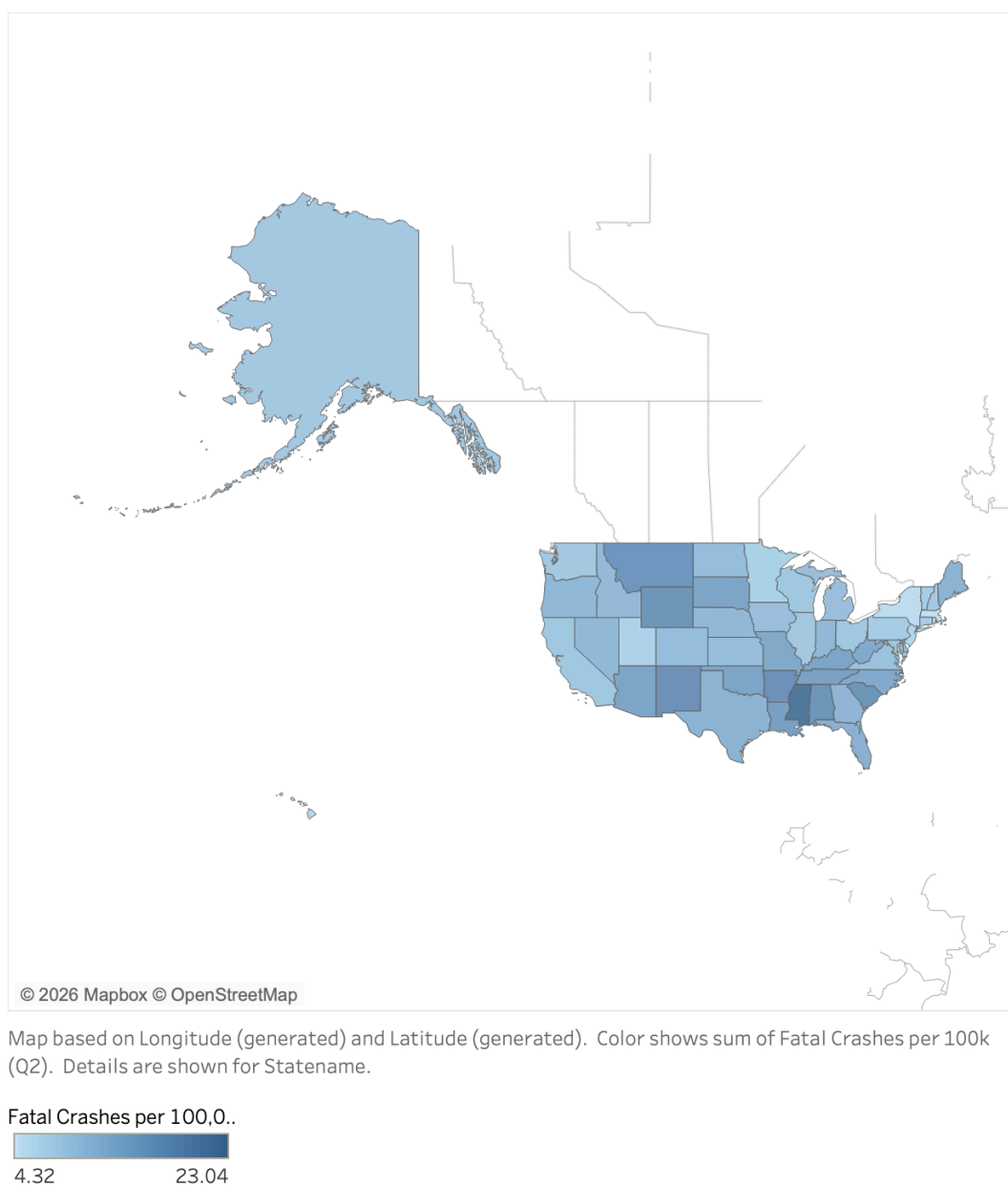
Figure 1: Fatal Crash Patterns by Day and Hour, 2024



Four of the five highest-count day-hour combinations occurred on Saturday, with the highest occurring between 8:00 and 8:59 PM. This concentration suggests that weekend evening hours may warrant further attention in traffic-safety planning.

Prototype 2: How does the burden of fatal crashes vary across states?

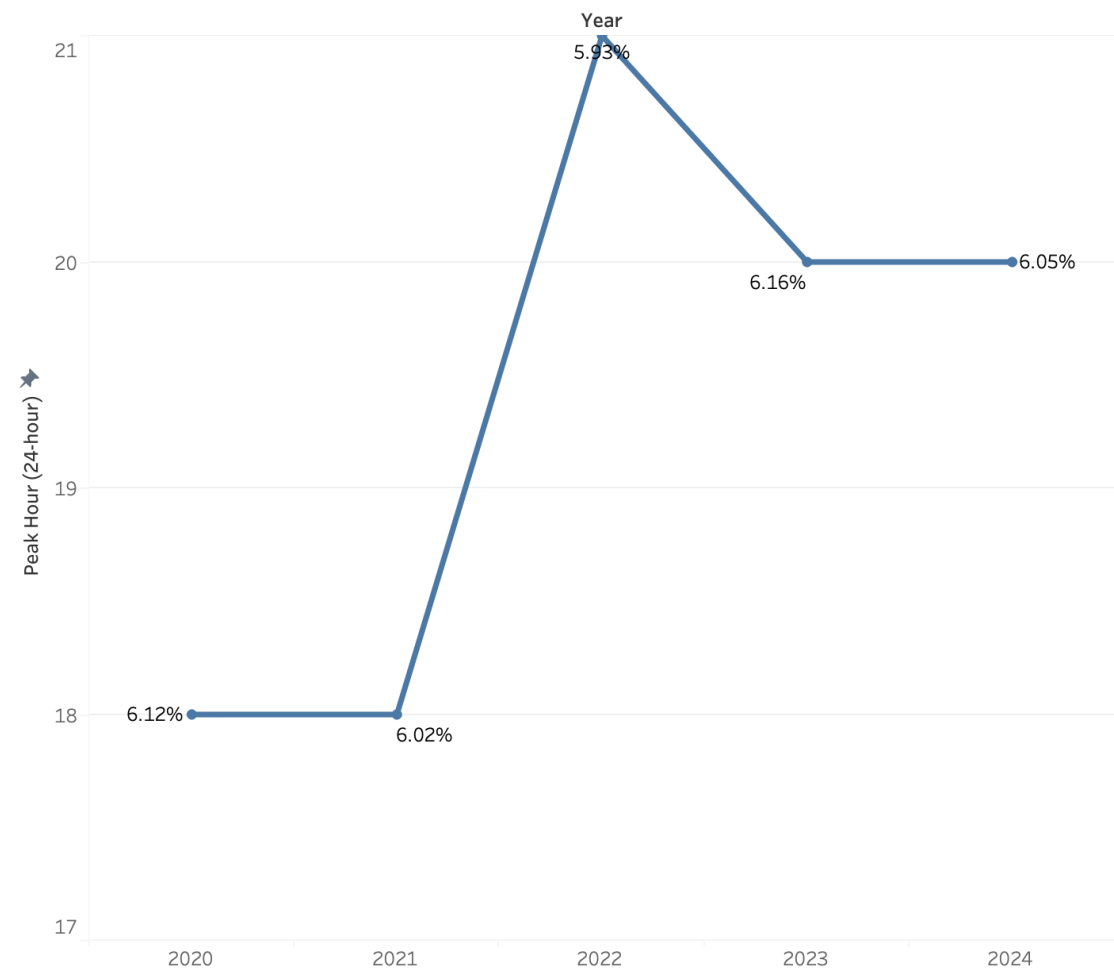
Figure 2: Fatal Crashes per 100,000 Residents by State, 2024



After accounting for differences in population size, Mississippi had the highest fatal crash rate in 2024 (23.04 per 100,000 residents), followed by New Mexico (17.74) and Arkansas (17.71). Across the 51 geographic areas included in the FARS data, fatal crash rates ranged from 4.32 to 23.04 per 100,000 residents, indicating substantial geographic variation.

Prototype 3: How have fatal crash patterns changed from 2020 to 2024?

Peak Hour for Fatal Crashes by Year, 2020–2024



The trend of Hour for Year. The marks are labeled by Percent Of Year as an attribute.

In 2020 and 2021, the peak hour for fatal crashes was from 6:00–6:59 PM. In 2022, it shifted to 9:00–9:59 PM. In 2023 and 2024, the peak hour for fatal crashes changed to 8:00–8:59 PM. Despite changes in the peak hour, the share of fatal crashes occurring during the peak hour remained relatively stable, ranging from 5.93% to 6.16% across 2020–2024.

4. Semester Portfolio Plan

The portfolio will begin with fatal crashes because they represent the most severe crash outcomes and provide a clear policy focus for transportation safety agencies. As the analysis develops, I will incorporate traffic-exposure measures such as VMT to account for differences in traffic exposure and allow more meaningful comparisons across states. If comparable state-level data on total reported crashes are available, I may also examine fatal crashes relative to the broader crash burden.

5. AI Use and Verification

I used ChatGPT to brainstorm research questions, evaluate visualization alternatives, explain Python and Tableau code, troubleshoot data-processing steps, and refine the presentation of findings.

For example, AI initially suggested using heatmap for five-year timing analysis. I revised this direction after testing the visualizations in Tableau and selected a line chart showing each year's peak fatal-crash hour because it more directly answers the research question.

I verified AI-assisted claims by inspecting the original FARS accident.csv files, checking variable definitions and coding, running the analysis independently in Python, and comparing the

resulting counts with the Tableau visualizations. I also revised the research questions based on the variables in the dataset instead of AI's ideas.

The final data cleaning, aggregation, Tableau construction, interpretation, and design decisions were reviewed directly against the downloaded data. AI was used as an analytical and coding assistant, not as a data source.